AI-Assisted Development Challenge for GET selection

Build a Mini AI Business Intelligence Assistant

Objective

Build a small application that allows a business user to ask questions about sales data using natural language and receive answers supported by actual database queries.

You may use AI coding assistants such as ChatGPT, Claude, Cursor, GitHub Copilot, or equivalent tools.

We are evaluating your ability to build a working application, understand the code, solve problems, and make sound technical decisions. You are not expected to build a production-ready system.

Problem Statement

You are provided with a CSV file containing sales transactions. (Attached below for your reference)

Build an application that enables users to ask questions such as:

1. What is the total revenue?
2. Which product generated the highest revenue?
3. What is the monthly revenue trend?
4. Which region generated the highest sales?
5. How did revenue change between two months?

Functional Requirements

1. Data Management

- Load the supplied CSV into a database.
- Handle missing or invalid records appropriately.
- Explain how the data is stored and queried.

2. Backend

- Build an API that accepts a user's question.
- Process the question and return a structured answer.
- Handle invalid requests and errors.

3. Analytics

- Support at least four of the five example questions.
- Calculate results using database queries or equivalent deterministic logic.
- Display the supporting figures used to produce the answer.

4. AI Integration

- Use an LLM or an AI-assisted interpretation mechanism.
- If using text-to-SQL, validate the generated query before execution.
- Do not allow destructive database operations.
- Explain how you prevent unsupported or incorrect answers.

5. User Interface

- Provide a simple interface to enter questions and view results.
- Display the answer and relevant supporting data.

6. Testing

- Include at least five test cases.
- Test valid questions, invalid questions, and at least one edge case.
- Document known limitations.

Deliverables

Submit:

1. Source code in a Git repository.
2. README with setup and execution instructions.
3. A short architecture diagram or explanation.
4. Automated tests.
5. A short document explaining:
    - Which AI tools you used.
    - How you used AI during development.
    - Which problems you encountered.
    - How you debugged them.
    - What you would improve with another day of work.

Time Limit

Target effort: 3–4 hours.

You may submit incomplete functionality, if necessary, provided you clearly explain what works, what does not, and why.

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Evaluation

Your submission will be assessed on:

- Working functionality.
- Programming fundamentals.
- Backend and database design.
- Correctness and reliability.
- Appropriate use of AI.
- Testing and error handling.
- Your ability to explain and modify your implementation.

There is no requirement to build an elaborate interface or use a particular AI framework.

Material for learning:

https://www.youtube.com/watch?v=rV3HJ4LEZ7k

https://www.deeplearning.ai/courses/agentic-ai

Concepts on which you will be interviewed:

Agentic AI, Generative AI, RAG, Vectorless RAG

Agentic Frameworks: Langchain, Google ADK, Dify

Please find the CSV file containing sales transactions. (Excel file of the same is also attached below for your reference)

Transaction_Id	Order_date	Product	Category	Region	Sales_channel	Quantity	Unit_price	Discount_amount	Status
TXN0001	03-01-2026	Laptop Stand	Accessories	North	Online	10	1200	0	Completed
TXN0002	05-01-2026	Wireless Mouse	Accessories	South	Retail	25	650	0	Completed
TXN0003	08-01-2026	Office Chair	Furniture	West	Online	4	7800	0	Completed
TXN0004	12-01-2026	Monitor	Electronics	East	Retail	6	9500	500	Completed
TXN0005	15-01-2026	Keyboard	Accessories	North	Online	18	900	0	Completed
TXN0006	18-01-2026	Desk	Furniture	South	Retail	3	12500	0	Completed
TXN0007	21-01-2026	USB-C Hub	Accessories	West	Online	12	1800	0	Completed
TXN0008	25-01-2026	Monitor	Electronics	North	Online	2	9500	0	Completed
TXN0009	02-02-2026	Laptop Stand	Accessories	East	Retail	14	1200	0	Completed
TXN0010	04-02-2026	Wireless Mouse	Accessories	West	Online	30	650	0	Completed
TXN0011	07-02-2026	Office Chair	Furniture	South	Online	5	7800	0	Completed
TXN0012	10-02-2026	Monitor	Electronics	North	Retail	8	9500	1000	Completed
TXN0013	14-02-2026	Keyboard	Accessories	East	Online	20	900	0	Completed
TXN0014	17-02-2026	Desk	Furniture	West	Retail	2	12500	0	Completed
TXN0015	21-02-2026	USB-C Hub	Accessories	South	Online	15	1800	0	Completed
TXN0016	25-02-2026	Wireless Mouse	Accessories	North	Retail	22	650	0	Completed
TXN0017	01-03-2026	Laptop Stand	Accessories	South	Online	16	1200	0	Completed
TXN0018	04-03-2026	Wireless Mouse	Accessories	East	Retail	28	650	0	Completed
TXN0019	07-03-2026	Office Chair	Furniture	North	Online	3	7800	0	Completed
TXN0020	10-03-2026	Monitor	Electronics	West	Online	5	9500	0	Completed
TXN0021	13-03-2026	Keyboard	Accessories	South	Retail	24	900	0	Completed

TXN0022	16-03-2026	Desk	Furniture	East	Online	4	12500	2500	Completed
TXN0023	20-03-2026	USB-C Hub	Accessories	North	Retail	10	1800	0	Completed
TXN0024	24-03-2026	Monitor	Electronics	South	Online	3	9500	0	Completed
TXN0025	02-04-2026	Laptop Stand	Accessories	West	Online	12	1200	0	Completed
TXN0026	05-04-2026	Wireless Mouse	Accessories	North	Retail	35	650	0	Completed
TXN0027	08-04-2026	Office Chair	Furniture	East	Online	4	7800	0	Completed
TXN0028	11-04-2026	Monitor	Electronics	South	Retail	7	9500	500	Completed
TXN0029	14-04-2026	Keyboard	Accessories	West	Online	21	900	0	Completed
TXN0030	17-04-2026	Desk	Furniture	North	Retail	3	12500	0	Completed
TXN0031	20-04-2026	USB-C Hub	Accessories	East	Online	18	1800	0	Completed
TXN0032	23-04-2026	Wireless Mouse	Accessories	South	Online	27	650	0	Completed
TXN0033	26-04-2026	Monitor	Electronics	West	Retail	2	9500	0	Returned
TXN0034	28-04-2026	Keyboard	Accessories	East	Online	0	900	0	Completed
TXN0035	02-05-2026	Laptop Stand	Accessories	North	Online	15	1200	0	Completed
TXN0036	05-05-2026	Wireless Mouse	Accessories	South	Retail	32	650	0	Completed
TXN0037	08-05-2026	Office Chair	Furniture	West	Online	6	7800	0	Completed
TXN0038	11-05-2026	Monitor	Electronics	East	Retail	9	9500	0	Completed
TXN0039	14-05-2026	Keyboard	Accessories	North	Online	26	900	0	Completed
TXN0040	17-05-2026	Desk	Furniture	South	Retail	4	12500	0	Completed
TXN0041	20-05-2026	USB-C Hub	Accessories	West	Online	20	1800	0	Completed
TXN0042	24-05-2026	Monitor	Electronics	North	Online	4	9500	0	Completed
TXN0043	03-06-2026	Laptop Stand	Accessories	East	Online	13	1200	0	Completed
TXN0044	06-06-2026	Wireless Mouse	Accessories	West	Retail	29	650	0	Completed
TXN0045	09-06-2026	Office Chair	Furniture	South	Online	5	7800	0	Completed

TXN0046	12-06-2026	Monitor	Electronics	North	Retail	6	9500	0	Completed
TXN0047	15-06-2026	Keyboard	Accessories	West	Online	19	900	0	Completed
TXN0048	18-06-2026	Desk	Furniture	East	Retail	2	12500	0	Completed
TXN0049	21-06-2026	USB-C Hub	Accessories	South	Online	16	1800	0	Completed
TXN0050	25-06-2026	Keyboard	Accessories	North	Retail	11	900	0	Completed
TXN0051	28-06-2026	Laptop Stand	Accessories	East	Online	5		0	Completed

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